Eye Diagrams

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1 Introduction

An eye diagram is a visual representation of a digital signal, created by overlaying multiple signal segments aligned with the symbol period, forming a pattern resembling a human eye. The key characteristics of an eye diagram include the eye opening, which reflects signal clarity where a wide opening indicates low noise and minimal distortion, while a closed or blurry eye suggests issues like noise and intersymbol interference. The vertical height of the eye shows the signal amplitude and noise margin, indicating how robust the signal is against voltage noise. The horizontal width relates to timing jitter and synchronization errors, revealing the timing accuracy of the signal transitions. Thick or blurred lines in the diagram point to increased noise or crosstalk. Additionally, the eye crossing points indicate the optimal sampling thresholds for receivers to decide bits accurately.

To make the eye diagrams, a sine wave of 2 [kHz] will be sent to the input of a sequence generator. That output will be sent to the input of a tunable low pass filter (LPF). The output will be the eye diagram. For more testing, a 100 [kHz] channel filter will be used to test the response in different scenarios. In order to see the eye diagrams more clearly, the persistence setting on the oscilloscope will be turned up. This setting makes previous measurements stay on screen for some time to see the overall shape better.

2 Data and Analysis

Figure 1 shows the eye diagram when using a 2[kHz] sequence generator for the input going through channel 1.



Figure 1: Generated sequence in green and resulting eye diagram in yellow

Figures 2 and 3 show the result of the same input going through channels 2 and 3 respectively.

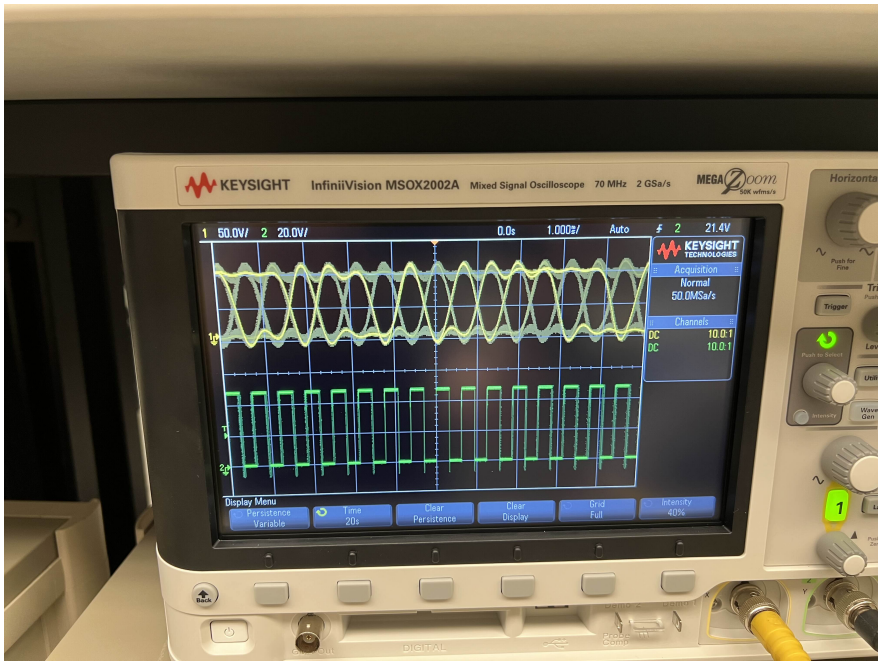


Figure 2: Eye diagram through channel 2

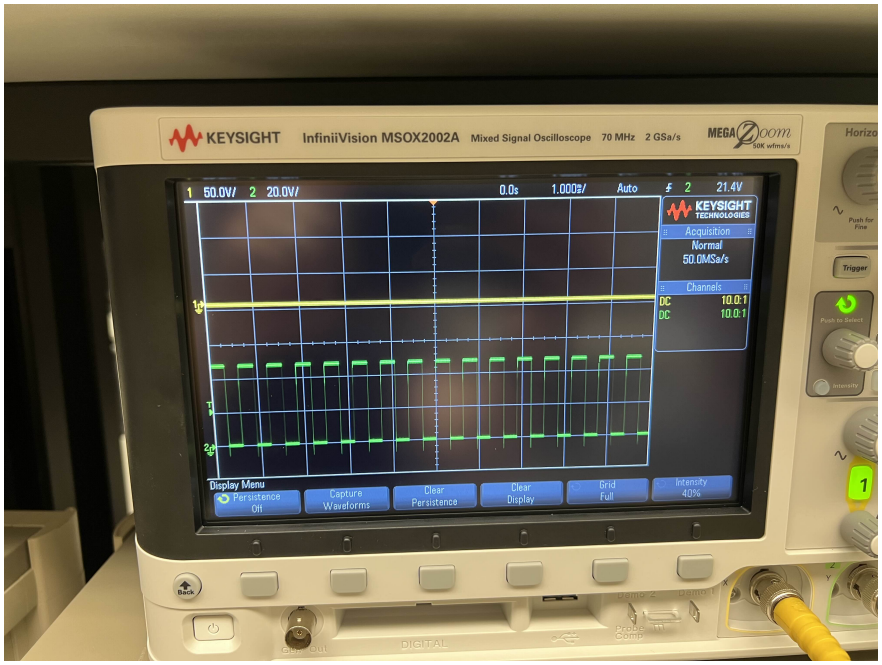


Figure 3: Eye diagram through channel 3

Channel 3 in this experiment consistently produced a null signal. It is believed that the modulus was not performing properly, meaning channel 3 will be ignored from here forward. Comparing channel 1 and 2, there was a slight amplitude change from channel 1 to 2. For channel 4, a direct connection to the tunable LPF will be made with a 1[kHz] sine wave. The resulting eye diagram is seen in Figure 3.

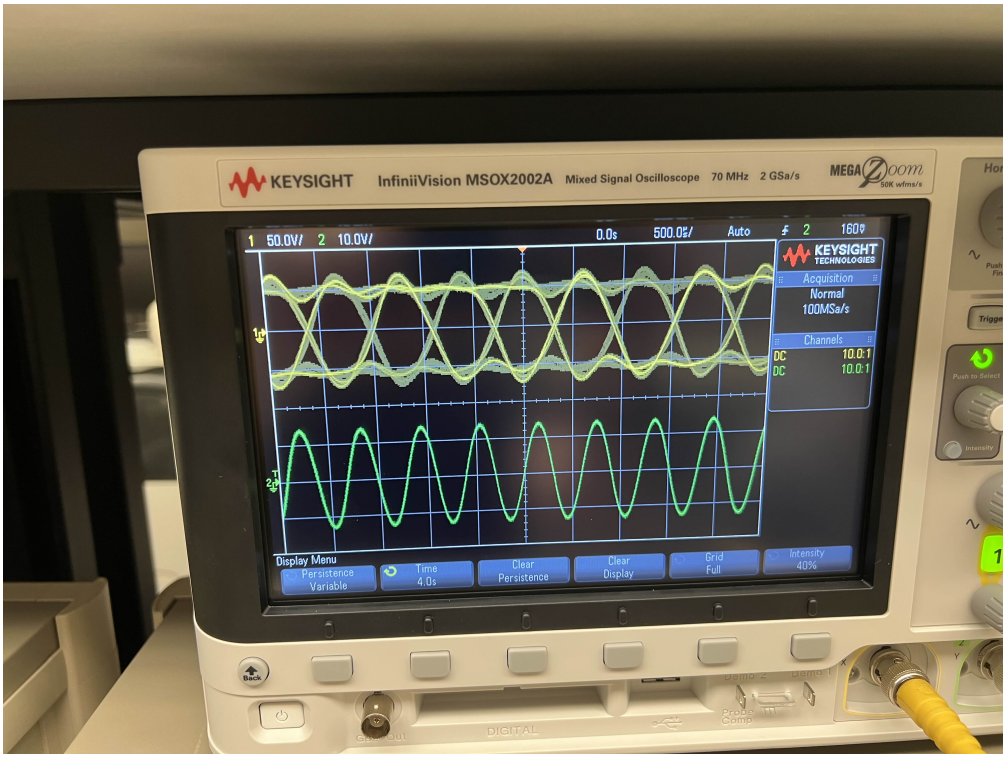


Figure 4: Eye diagram through channel 4

The eye diagram generated has more noticeable changes from 1 and 2. The eye diagram has a similar height indicating similar SNR values. The wider eye means there are less distortions compared to channels 1 and 2 which is expected with a more direct connection. Eye diagrams are also used to assess the maximum data rate of a channel. Once the eye becomes distorted, the receiver will have a harder time distinguishing between values. The max data rate results for channels 1, 2 and 4 are seen in figures 5, 6 and 7 respectively.

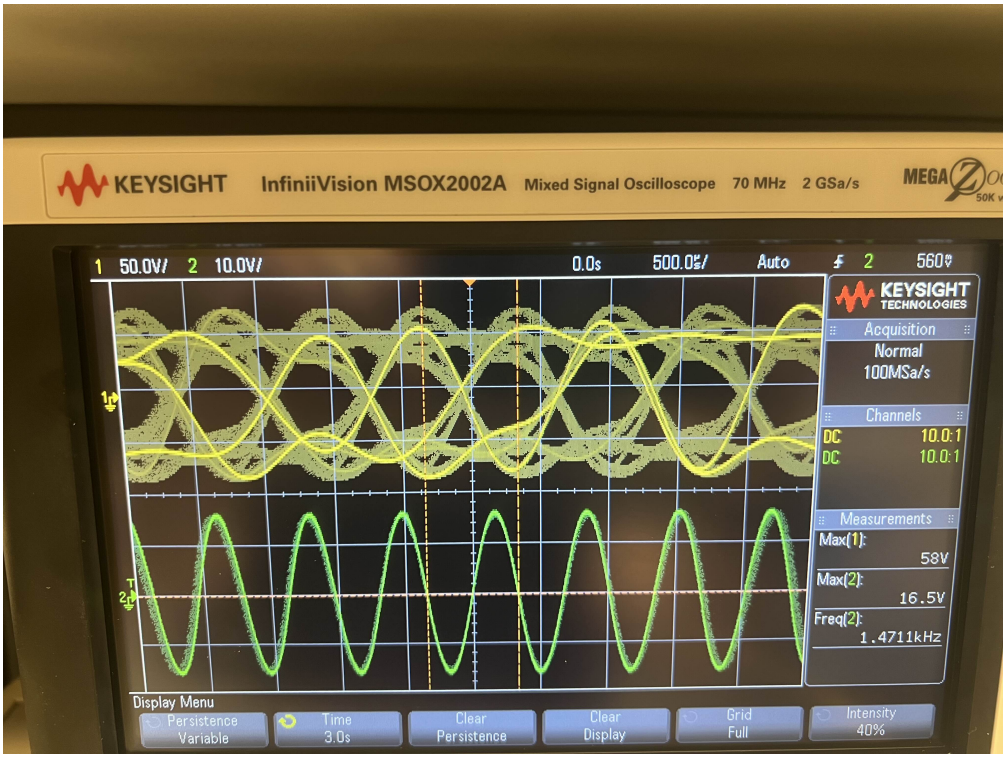


Figure 5: Eye diagram for channel 1 with too high of a data rate

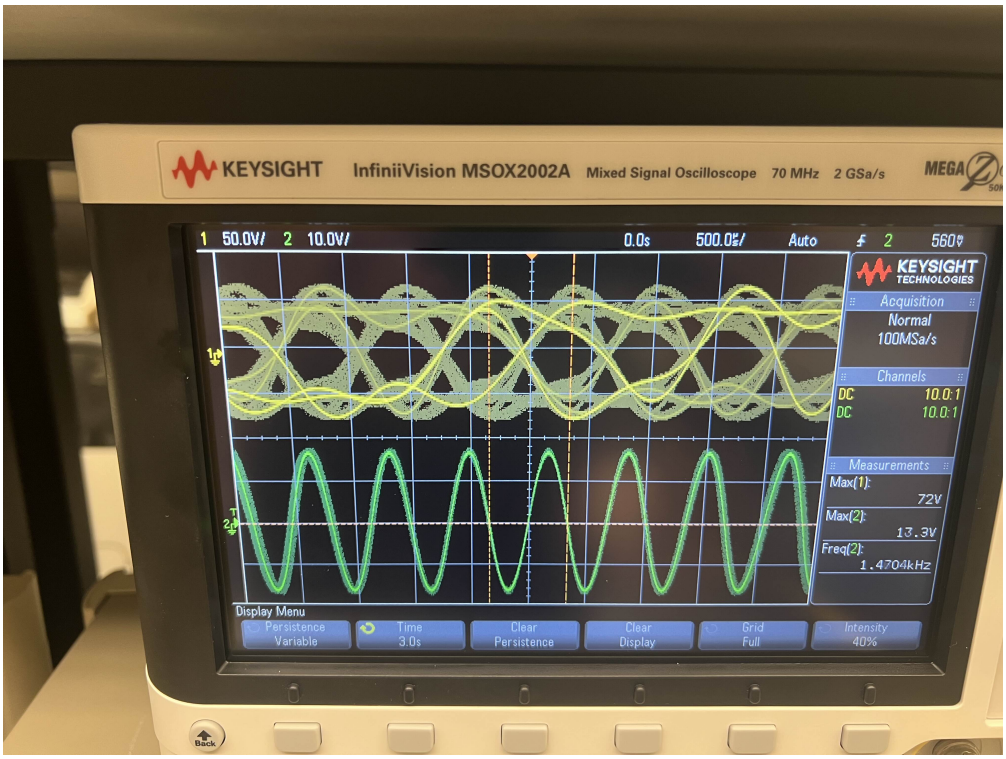


Figure 6: Eye diagram for channel 2 with too high of a data rate

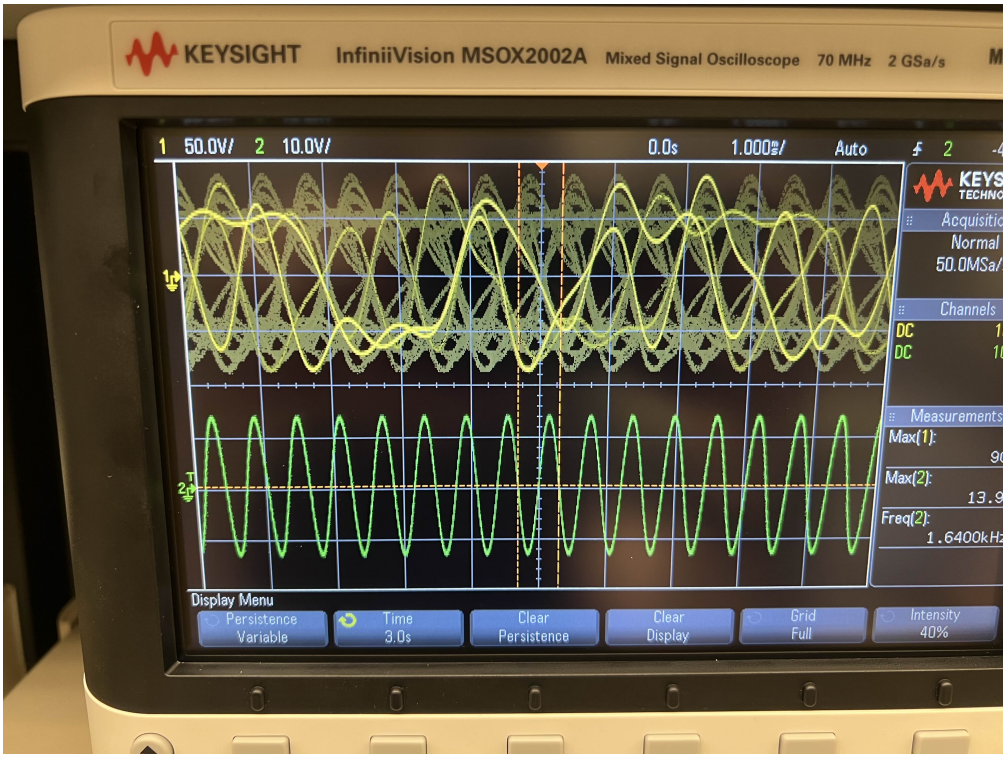


Figure 7: Eye diagram for channel 4 with too high of a data rate

The characteristics of an eye diagram with too high a data rate are clear intersections between symbols resulting in unclear eyes. Channel 1 and 2 have a similar max data rate being 1.471 [kHz] for channel 1 and 1.470 [kHz] for channel 2. Channel 4 has a higher data rate given the direct connection being 1.64 [kHz].

Different data types can also effect eye diagrams and their stability. For example, when using a square wave of 2[kHz], the resulting eye diagram is found in Figure 8.

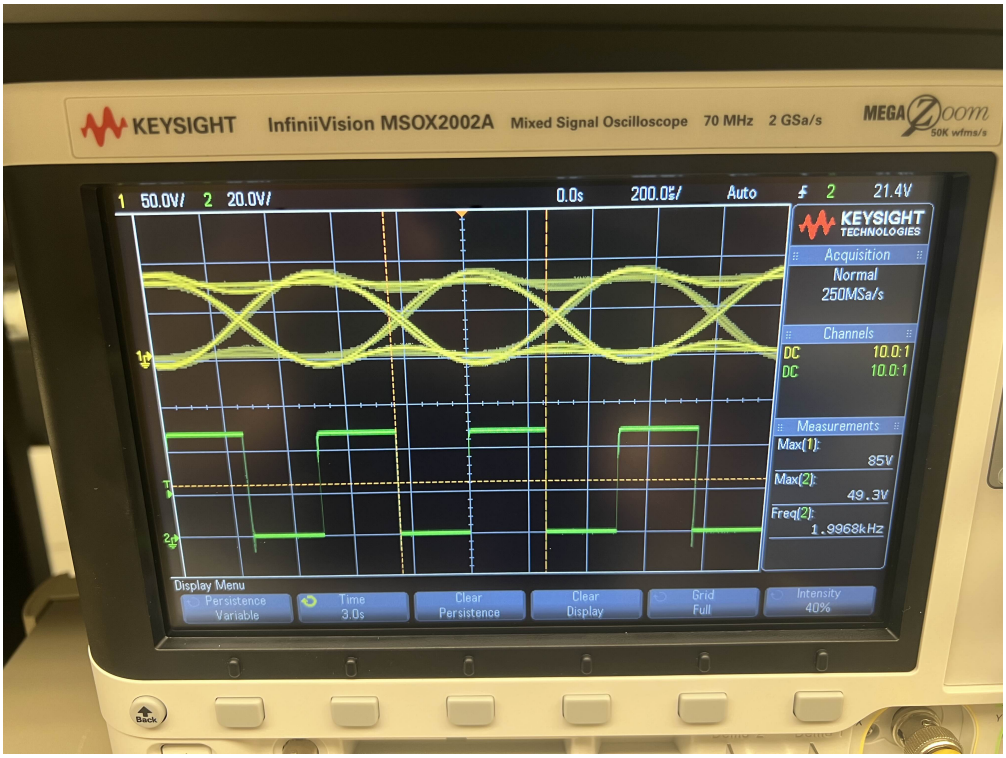


Figure 8: Eye diagram for channel 2 with a square wave input

This eye diagram is much clearer then before with less vertical hight but increased width for better timing. This is fairly intuitive as the incoming signal will be the same value for the width of the pulse. The resulting max data rate in shown in figure 9.

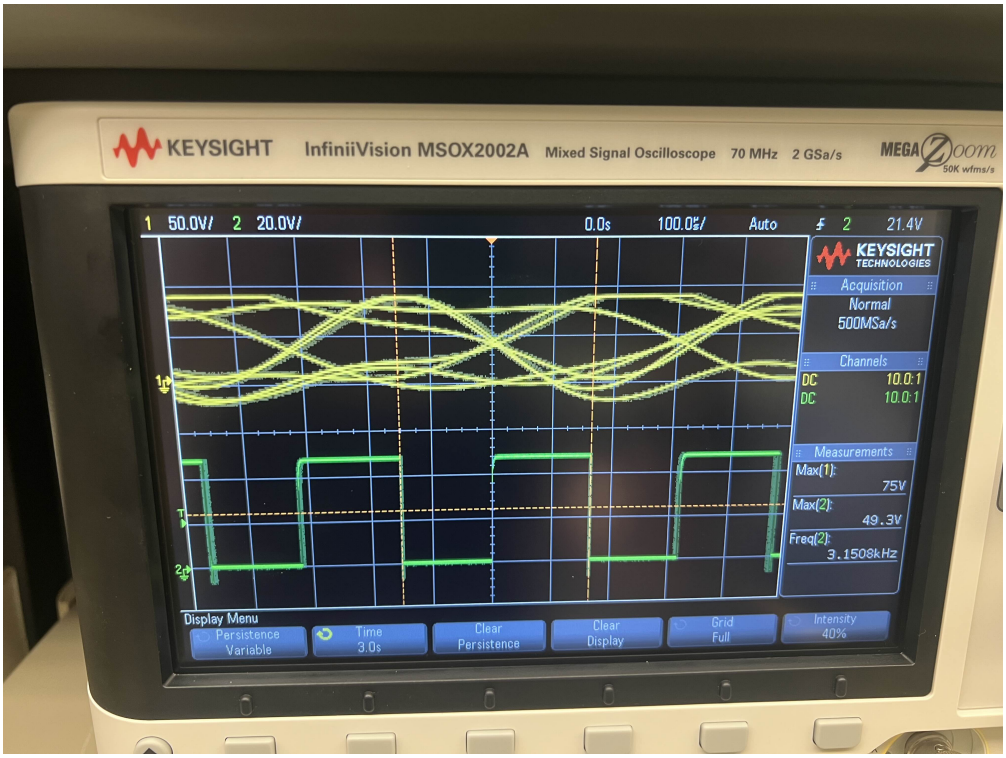


Figure 9: Eye diagram for channel 2 with too high of a bit rate

Certification: **I hereby certify that performance of the experiment above and the writing of this report was entirely by me (or by me and my lab partner). I understand that if this certification is false, I am in violation of Academic Honesty rules and may be subject to serious penalty in accordance with those rules.**

Signature Josiah Germain **Date** 11/20/25